

Venkata Rajesh Paidipala

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OH, Cincinnati (willing to relocate)

PROFESSIONAL SUMMARY:

- Network Engineer with hands-on experience in routing, switching, and **securing enterprise** and service provider networks using protocols like **OSPF, BGP, MPLS, VLANs, and VPNs**.
- Proficient in **network security and firewall configurations** implementing VPNs, ACLs, and IDS/IPS for secure data transmission and threat prevention.
- **Skilled in cloud networking** on **AWS and Azure**, including **VPC setup, IAM** policy management, and hybrid cloud integration for scalable infrastructure.
- Experienced in **automating network tasks using Python Ansible, and Terraform**, improving deployment speed, consistency, and reducing manual errors.
- Adept at using tools such as **Wireshark, PRTG, and SolarWinds** for **real-time monitoring**, traffic analysis, and performance optimization.
- **Strong team player** with a proactive approach **to problem-solving**, committed to building secure, high-performing, and resilient network systems.

TECHNICAL SKILLS:

Core Networking:

OSPF, BGP, EIGRP, RIP, MPLS, VLANs, VTP, STP/RSTP/MSTP, EtherChannel, QoS, VRF, DNS, DHCP, NTP, EtherChannel, NAT, SNMP, L2/L3 VPNs, Spine-Leaf Architecture.

Security:

Cisco ASA, Palo Alto, **Checkpoint Firewalls, VPNs, TACACS+, RADIUS, Zero Trust Architecture, ACLs, Vulnerability Management**

Cloud & Virtualization:

AWS (VPC, EC2, S3, IAM) Azure Networking, GCP Networking, VMware, Hyper-V.

Automation & Scripting:

Python, Ansible, Terraform, **Bash, PowerShell**

Tools:

GitHub Actions, GitLab CI, Git, Agile/Scrum

Monitoring & Testing Tools:

Wireshark, SolarWinds, PRTG, **Spirent**, Cisco Packet Tracer, Nmap, Metasploit

Operating Systems:

Linux (**Ubuntu, CentOS, Red Hat**), Shell Scripting

Experience:

Company: Tejas Networks

Sep 2022 – Dec 2023

Role: Research and Development PE Engineer

Technologies: MPLS, OSPF, BGP, RSVP-TE, L2VPN, IPV4,IPV6, SD-WAN, VLANs, Python, Linux, Ansible, NMS/EMS, Spirent, IXIA, ARP, IS-IS,RIP,etherChannel,NAT(Static,Dynamic,PAT).

Responsibilities:

- Conducted detailed manual and automated testing of advanced routing protocols (**BGP, EIGRP, RSVP-TE**) across multi-vendor environments.
- Validated L2/L3 VPNs, VPLS, and **MPLS** configurations for secure and scalable deployments in enterprise and **SD-WAN** setups.
- Integrated Tejas equipment into existing service provider networks, coordinating interoperability with **Juniper, and Cisco platforms**.
- Managed **VLANs, VPNs, firewall** configurations and network monitoring via **EMS/NMS** tools.
- Participated in **Vulnerability Assessment and Penetration Testing (VAPT)** for the TJ1400P-H switch supporting security compliance.
- Automated test scenarios and configurations using **Python and Ansible**, improving testing **efficiency and accuracy**.
- Performed network troubleshooting and testing **using ICMP, UDP, and TCP** protocols to identify connectivity issues, measure network latency, and verify reliable data transmission across various network layers.
- Familiar and hands-on with traffic generation tools like **Spirent and Ixia**, including **1G, 10G, and 100G** testers for performance validation and network stress testing.

Company: Capgemini

Nov 2020 – August 2021

Role: Network Analyst

Technologies:

OSPF, BGP, VLANs, QoS, Wireshark, PRTG, SolarWinds, SNMP, Python, PowerShell

Responsibilities:

- Configured **L2 switches** and routing protocols (**OSPF, BGP**) to optimize enterprise network performance.
- Analyzed **real-time traffic using Wireshark and hping** to detect anomalies and evaluate **DDoS/DoS** resilience.
- Delivered rapid troubleshooting support in production environments, **minimizing downtime** and ensuring uptime targets.
- Developed custom scripts in **Python and PowerShell for routine automation** and network diagnostics.
- Developed test scenarios for various network configurations, including **VLANs and QoS** to validate their performance, security, and fault tolerance.
- Configured and managed **Virtual Chassis technology**, combining multiple network devices into a **unified logical unit** to enhance scalability, improve redundancy, and streamline network management.

Education:

Master of Science in Information Technology

Jan 2024 – April 2025

University of Cincinnati

GPA : 3.93 Scale of 4.0

Projects:

1. Title: Comprehensive Network Optimization and Secure SD-WAN Deployment. -Tejas Netw

Technologies: OSPF, BGP, VLANs, ACLs, MPLS, SD-WAN, VPN, QoS, Network Security, Wireshark, Network Monitoring Tools.

Description:

- Designed and deployed dynamic routing (OSPF, RIP, EIGRP, BGP) for a **robust and redundant infrastructure**.
- Segmented **network traffic** using **VLANs** and implemented **ACLs** for access control.
- Established encrypted **VPN tunnels** and deployed SD-WAN to improve **multi-site connectivity**.
- Prioritized VoIP and video traffic with **QoS** and analyzed performance using Wireshark and other tools.

2. Layer 2 DoS Simulation and Mitigation via Automated MAC Flooding - Tejas Networks

Technologies: Python, Scapy, ARP, MAC Spoofing, Port Security, VLANs, Wireshark, Network Monitoring

- Simulated a **MAC flooding DoS attack** using Python and **Scapy from Kali Linux** to overflow a Tejas TJ1400 switch's MAC table.
- Monitored the attack using Wireshark on a mirror port to observe unicast **flooding and unusual ARP behavior**.
- Applied **port security** settings on the switch to **limit MAC addresses** and prevent further flooding.
- Automated the attack process and documented **mitigation strategies** security improvements.

3. Title: Zero Trust Security Architecture on AWS - Univ of Cincinnati

Technologies Used: AWS IAM, VPC, Security Groups, NACLs, AWS WAF, Guard Duty, Terraform.

Description:

- Built a Zero Trust model on **AWS with strict access policies, MFA**, and segmented networks.
- Applied **VPC designs** with secure subnetting, **WAF rules**, and centralized logging via **CloudTrail**.
- Deployed Guard Duty for threat detection and automated secure infrastructure provisioning with Terraform.

4. Message Security Simulator with Classic Ciphers in Python" - Univ of Cincinnati

Tools/Technologies: Python, PyCharm, Cipher Text, Morse Code, Basic Cryptography

- Designed a Python-based application to **encode and decode messages** using classic cryptographic methods including **Morse code** and **Caesar Cipher**.
- Gained insight into **how attackers exploit weak encryption** and how to strengthen it using layered encoding and secure key exchanges.
- Practiced secure development using **PyCharm**, including input validation and exception handling to prevent misuse.